

Program 5:

CI Integration by Running Jenkins in a Docker Container

Project Structure

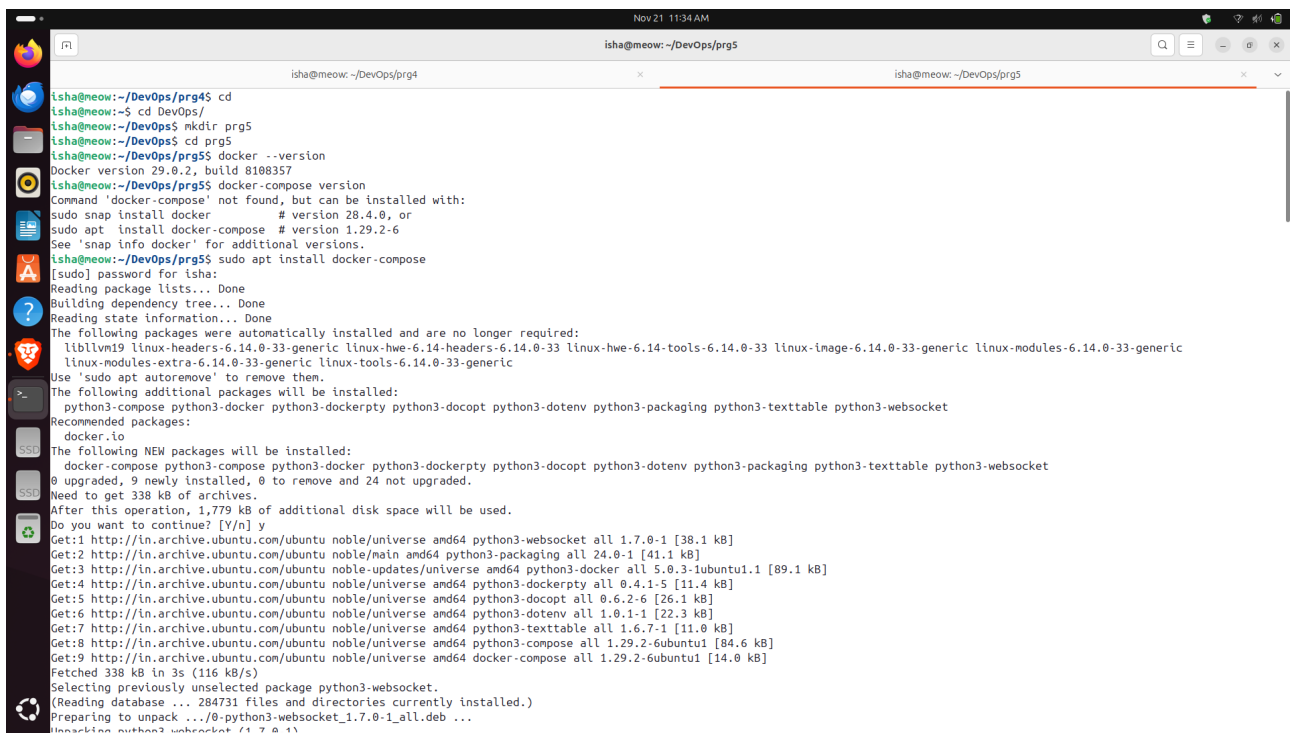
```
program-5/  
├── Dockerfile  
└── docker-compose.yml
```

Step 1: Start Docker Service

```
docker --version  
docker compose version
```

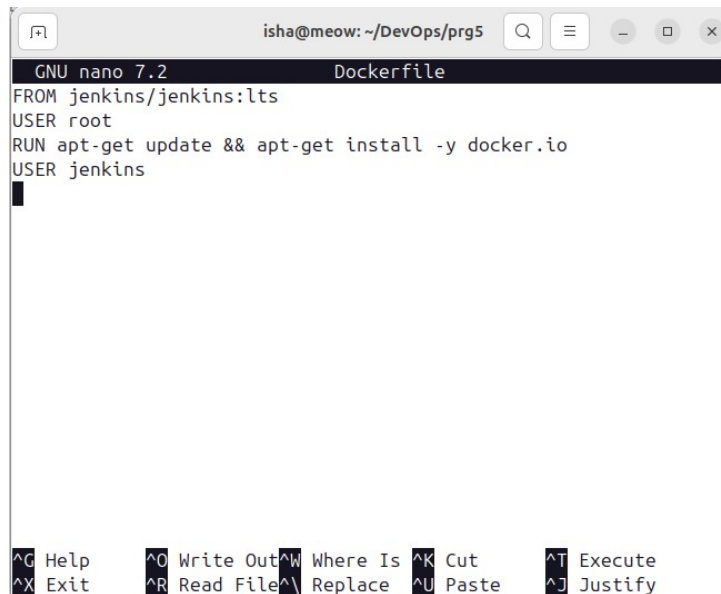
Step 2: Create Project Directory

```
mkdir program-5  
cd program-5
```



```
Nov 21 11:34 AM  
isha@meow: ~/DevOps/prg5  
isha@meow:~/DevOps/prg4$ cd  
isha@meow:~$ cd DevOps/  
isha@meow:~/DevOps$ mkdir prg5  
isha@meow:~/DevOps$ cd prg5  
isha@meow:~/DevOps/prg5$ docker --version  
Docker version 29.0.2, build 8108357  
isha@meow:~/DevOps/prg5$ docker-compose version  
Command 'docker-compose' not found, but can be installed with:  
sudo snap install docker          # version 28.4.0, or  
sudo apt install docker-compose  # version 1.29.2-6  
See 'snap info docker' for additional versions.  
isha@meow:~/DevOps/prg5$ sudo apt install docker-compose  
[sudo] password for isha:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following packages were automatically installed and are no longer required:  
  libllvm19 linux-headers-6.14.0-33-generic linux-hwe-6.14-headers-6.14.0-33 linux-hwe-6.14-tools-6.14.0-33 linux-image-6.14.0-33-generic linux-modules-6.14.0-33-generic  
Use 'sudo apt autoremove' to remove them.  
The following additional packages will be installed:  
  python3-compose python3-docker python3-dockerpty python3-docopt python3-dotenv python3-packaging python3-texttable python3-websocket  
Recommended packages:  
  docker.io  
The following NEW packages will be installed:  
  docker-compose python3-compose python3-docker python3-dockerpty python3-docopt python3-dotenv python3-packaging python3-texttable python3-websocket  
0 upgraded, 9 newly installed, 0 to remove and 24 not upgraded.  
Need to get 338 kB of archives.  
After this operation, 1,779 kB of additional disk space will be used.  
Do you want to continue? [Y/n] y  
Get:1 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 python3-websocket all 1.7.0-1 [38.1 kB]  
Get:2 http://in.archive.ubuntu.com/ubuntu noble/main amd64 python3-packaging all 24.0-1 [41.1 kB]  
Get:3 http://in.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-docker all 5.0.3-1ubuntu1.1 [89.1 kB]  
Get:4 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 python3-dockerpty all 0.4.1-5 [11.4 kB]  
Get:5 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 python3-docopt all 0.6.2-6 [26.1 kB]  
Get:6 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 python3-dotenv all 1.0.1-1 [22.3 kB]  
Get:7 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 python3-texttable all 1.6.7-1 [11.0 kB]  
Get:8 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 python3-compose all 1.29.2-6ubuntu1 [84.6 kB]  
Get:9 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 docker-compose all 1.29.2-6ubuntu1 [14.0 kB]  
Fetched 338 kB in 3s (116 kB/s)  
Selecting previously unselected package python3-websocket.  
(Reading database ... 284731 files and directories currently installed.)  
Preparing to unpack .../0-python3-websocket_1.7.0-1_all.deb ...  
Unpacking python3-websocket (1.7.0-1)
```

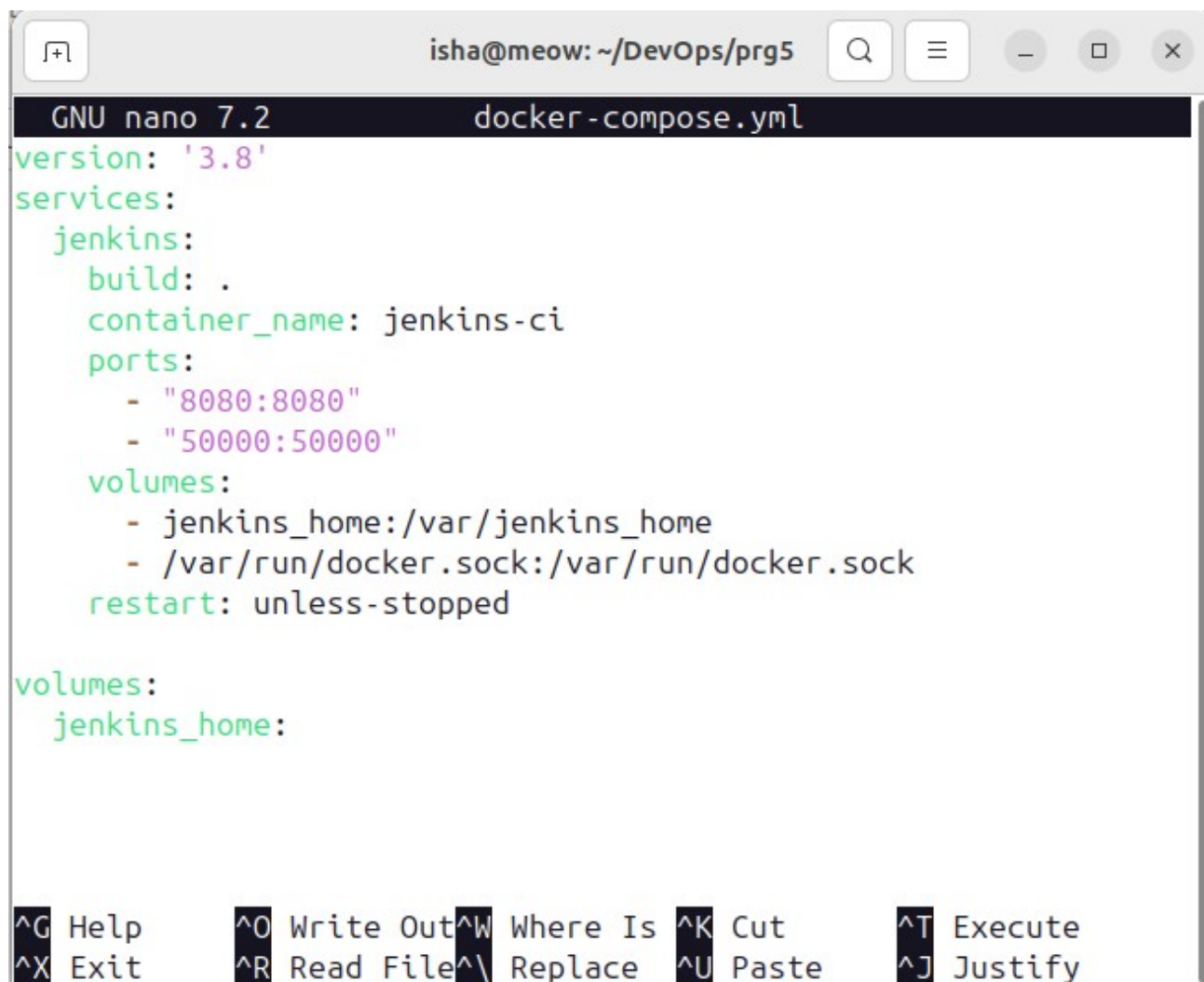
Step 3: Dockerfile Creation



```
GNU nano 7.2 Dockerfile
FROM jenkins/jenkins:lts
USER root
RUN apt-get update && apt-get install -y docker.io
USER jenkins
```

Step 4: Docker Compose Configuration

nano docker-compose.yaml



```
GNU nano 7.2 docker-compose.yml
version: '3.8'
services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock
    restart: unless-stopped

volumes:
  jenkins_home:
```

Step 5: Run Jenkins Container

docker compose up -d

docker log jenkins-ci

```
Nov 21 11:34 AM
isha@meow: ~/DevOps/prg4
isha@meow: ~/DevOps/prg5
isha@meow: ~/DevOps/prg5

=> exporting to image
=> => exporting layers
=> => writing image sha256:88a2c068b40d350ed7559bc13036f513eb6aa0afa5c7ed3b0f826b472826fb90
=> => naming to docker.io/library/prg5-jenkins
=> => resolving provenance for metadata file
[+] Building 1/1
prg5-jenkins Built
isha@meow:~/DevOps/prg5$ docker compose up -d
WARN[0000] /home/isha/DevOps/prg5/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 3/3
  ✓ Network prg5_default Created
  ✓ Volume prg5_jenkins_home Created
  ✓ Container jenkins-ci Started
isha@meow:~/DevOps/prg5$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
5d064f427138   prg5-jenkins  "/usr/bin/tini -- /u..." 4 seconds ago  Up 4 seconds  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
21dd2ea69c33   prg4        "docker-entrypoint.s..." 10 minutes ago Up 10 minutes  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
lsted_jepsen
isha@meow:~/DevOps/prg5$ docker log jenkins
docker: unknown command: docker log

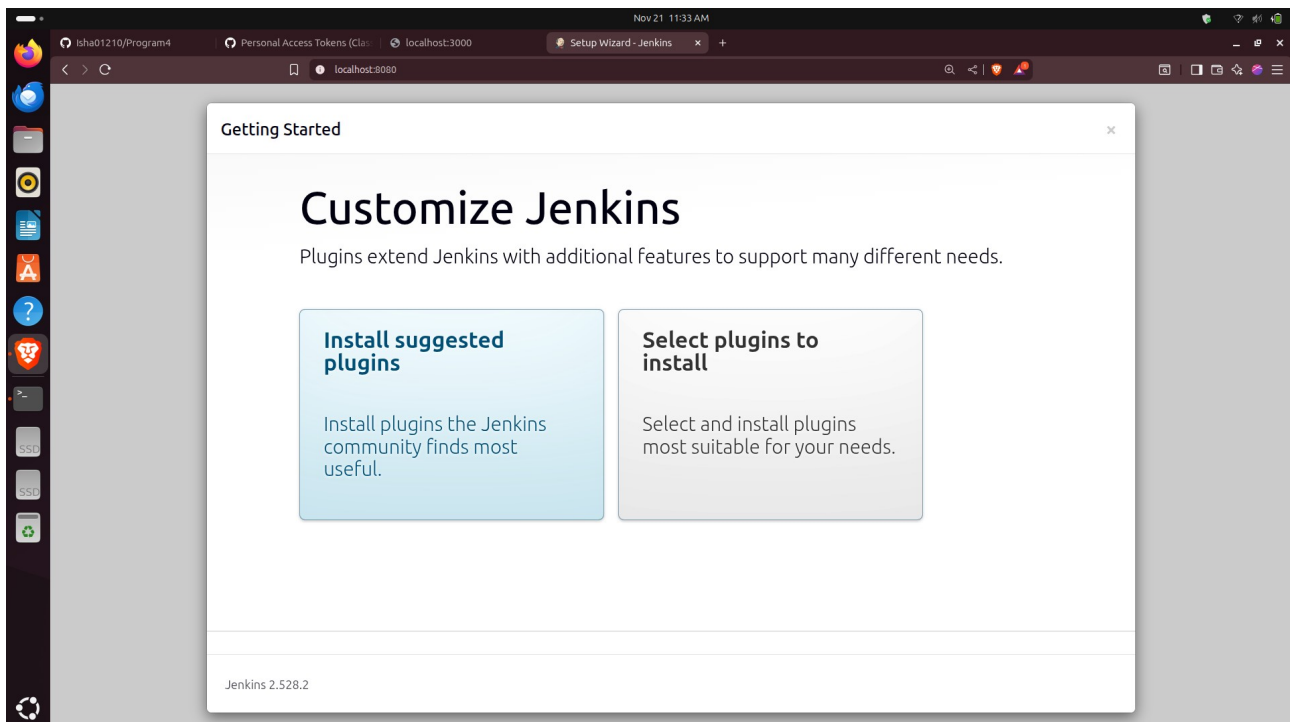
Run 'docker --help' for more information
isha@meow:~/DevOps/prg5$ docker logs jenkins
Error response from daemon: No such container: jenkins
isha@meow:~/DevOps/prg5$ docker logs jenkins:latest
Error response from daemon: No such container: jenkins:latest
isha@meow:~/DevOps/prg5$ docker logs prg5-jenkins
Error response from daemon: No such container: prg5-jenkins
isha@meow:~/DevOps/prg5$ docker logs
docker: 'docker logs' requires 1 argument

Usage: docker logs [OPTIONS] CONTAINER

Run 'docker logs --help' for more information
isha@meow:~/DevOps/prg5$ docker logs jenkins-ci
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2025-11-21 05:11:57.943+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2025-11-21 05:11:58.732+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2025-11-21 05:11:58.782+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.0.25; built: 2025-08-11T23:52:37.219Z; git: a862b76d8372e24205765182d9ae1d1d333ce2ea;
jvm 21.0.9+10-LTS
2025-11-21 05:11:59.259+0000 [id=1] INFO o.e.i.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.isp.JettyJspServlet
```

Step 6: Access Jenkins Web Interface

<http://localhost:8080>



Step 7: Cleanup

docker compose down

docker images

docker image rm program5-jenkins:latest

```
isha@neow:~/DevOps/prg5$ docker compose down
```

```
WARN[0000] /home/isha/DevOps/prg5/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
```

```
[+] down 2/2
```

```
✓ Container jenkins-ci Removed
```

```
0.4s
```